

CMPG 325 — COMPUTER NETWORKS

FINAL TECHNICAL REPORT

Molemane Community Newspaper Network Implementation

Secure Remote Management Using SSH

Project ID	CMPG325-2026-128
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Assigned networking challenge	SSH — secure device management plane
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1. Executive Summary

This report presents the completed Cisco Packet Tracer network for Molemane Community Newspaper, based on the Milestone 1 design and the Milestone 2 implementation. The network uses one internal boundary router (R1), one Layer 3 core switch (SW-CORE), and five access switches for Administration, Editorial, Sales, Design and the Boardroom. A dedicated server and management workstation support shared internal resources and secure administration.

Eight VLANs are implemented using the assigned 172.30.84.0/23 address block and a VLSM addressing plan. SW-CORE performs inter-VLAN routing through switch virtual interfaces (SVIs). VLAN 70 is configured as a reservation for the CR11 voice requirement; no VoIP handsets have been installed. R1 has no connection to an ISP or the Internet.

The assigned SSH challenge was implemented on R1 and all six switches. Final-stage corrections ensured that the available VTY lines on the five access switches accept SSH-only logins using local authentication. Remote SSH access was retested from PC-MANAGEMENT, with administrator privilege level 15 confirmed on all seven devices; Telnet attempts were rejected.

Final connectivity testing verified communication between departmental VLANs and the server. A separate wireless boardroom laptop was added to test the existing access point. It was configured in VLAN 50, connected using WPA2-PSK, and successfully reached its VLAN gateway and the server. The report preserves Milestone 2 context while clearly identifying final-stage tests and improvements. Successful pings verify IP reachability; they do not independently verify file-sharing, printing, or DNS application functions.

2. Introduction

2.1 Recap of Milestone 1

Milestone 1 established the client requirements, physical and logical network designs, and the VLSM addressing plan based on 172.30.84.0/23. The planned network separates departments by VLAN and includes dedicated wired and wireless boardroom access. The design also made provision for secure SSH administration and reserved VLAN 70 for future VoIP handsets under change request CR11.

R1 was designated as an internal boundary router without an Internet or ISP connection. SW-CORE was selected as the device responsible for inter-VLAN routing. The design and addressing documents remain part of the original milestone evidence in the GitHub portfolio.

2.2 Purpose of the Final Technical Report

The purpose of this report is to present the completed network implementation and explain how the final working Packet Tracer file compares with the implementation documented in Milestone 2. It records final network testing, SSH management improvements, wireless security changes, and remaining scope limits. The earlier milestone report is not replaced; this document provides a separate final-state account.

2.3 Structure of this Report

Section 3 describes the implemented topology, VLANs, addressing, trunks, SVIs, management subnet and router. Section 4 discusses the SSH challenge and verification. Section 5 records connectivity, SSH and trunk tests. Section 6 interprets the evidence, Section 7 documents troubleshooting and subsequent improvements, Section 8 concludes, and Section 9 lists references.

3. Network Implementation Overview

3.1 Final Topology

The completed topology retains the five access switches SW-ADMIN, SW-EDIT, SW-SALES, SW-DESIGN and SW-BOARD, all connected through SW-CORE. The server SRV-FILE-PRINT-DNS is attached to VLAN 60, and PC-MANAGEMENT uses VLAN 99 for SSH management. PC-ADMIN, PC-EDIT, PC-SALES and PC-DESIGN connect through their departmental access switches.

SW-BOARD supports the boardroom wired presentation device and BOARDROOM-WIRELESS-AP. A separate BOARDROOM-WIRELESS-LAPTOP was added during final testing. The wireless laptop uses Wireless0, IP address 172.30.84.131/28, and gateway 172.30.84.129. Its wireless access was secured with WPA2-PSK. This addition verifies the wireless requirement without changing the existing wired boardroom connection.

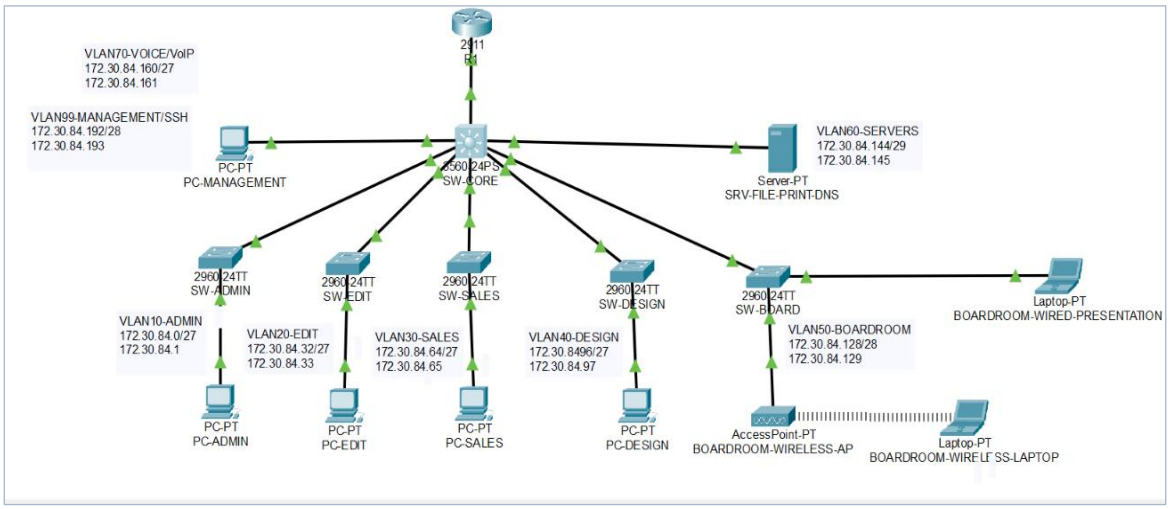


Figure 1 — Full network topology as built in the final Packet Tracer file

3.2 VLAN Allocation

The network contains eight purpose-specific VLANs, consistent with the Milestone 1 plan. Departmental traffic is segmented through VLANs 10–40, while VLAN 50 supports the boardroom, VLAN 60 contains the server, VLAN 70 is reserved for CR11 voice devices, and VLAN 99 carries device management traffic.

Table 1 — VLAN allocation

VLAN	Name	Purpose	Relevant access connection
10	ADMIN	Administration	SW-ADMIN; departmental access ports
20	EDITORIAL	Editorial / newsroom	SW-EDIT; departmental access ports
30	SALES	Advertising & sales	SW-SALES; departmental access ports
40	DESIGN	Design & pre-press	SW-DESIGN; departmental access ports
50	BOARDROOM	Wired presentation, AP and wireless laptop	SW-BOARD; wired and AP ports
60	SERVERS	Internal file / print / DNS server	SW-CORE; server access port
70	VOICE	Reserved for future CR11 VoIP handsets	No VoIP host access ports in use
99	MANAGEMENT	SSH management plane; trunk native VLAN	Switch SVIs and PC-MANAGEMENT

3.3 IP Addressing

VLSM was used to allocate subnet sizes appropriate to the anticipated host requirements. The four departmental VLANs and the reserved voice VLAN use /27 subnets; the boardroom and management VLANs use /28 subnets; the server VLAN uses /29; and the R1–SW-CORE point-to-point connection uses /30. The first usable address is configured as the SVI gateway for each VLAN.

Table 2 — IP addressing plan

VLAN / link	Subnet	Mask	Gateway / interface
10	172.30.84.0/27	255.255.255.224	172.30.84.1
20	172.30.84.32/27	255.255.255.224	172.30.84.33
30	172.30.84.64/27	255.255.255.224	172.30.84.65
40	172.30.84.96/27	255.255.255.224	172.30.84.97
50	172.30.84.128/28	255.255.255.240	172.30.84.129
60	172.30.84.144/29	255.255.255.248	172.30.84.145
70	172.30.84.160/27	255.255.255.224	172.30.84.161
99	172.30.84.192/28	255.255.255.240	172.30.84.193
R1 - SW-CORE	172.30.84.208/30	255.255.255.252	R1 .209 / SW-CORE .210

The previously reserved 172.30.84.208/30 block was used for the internal point-to-point link between R1 and SW-CORE. This does not represent a WAN or ISP connection. The new wireless laptop at 172.30.84.131 uses an available address within VLAN 50.

3.4 Trunk Configuration

Inter-switch connections use 802.1Q trunking, and the native VLAN is consistently set to VLAN 99 on each end of the configured trunks. This corrected the native VLAN mismatch found during implementation. On an access switch, the trunk carries the department-specific VLAN and VLAN 99; on SW-CORE, the uplinks were configured to carry the required VLANs.

Table 3 — Trunk configuration summary

Parameter	Configuration
Encapsulation	IEEE 802.1Q (dot1q)
Operating mode	Trunk
Native VLAN	99 on both sides
Access-switch allowed VLANs	Relevant department / boardroom VLAN plus VLAN 99
SW-CORE uplinks	Configured to transport required VLANs

3.5 Inter-VLAN Routing and SVI Configuration

SW-CORE operates as the Layer 3 gateway for the VLANs through switch virtual interfaces. Data-VLAN gateways are not duplicated on the access switches; their VLAN 99 SVIs provide management addresses only. The `ip routing` setting on SW-CORE allows traffic to be routed between active directly connected VLAN networks.

Table 4 — SVI configuration on SW-CORE

Interface	Purpose	IP address	Mask
Vlan10	ADMIN	172.30.84.1	255.255.255.224
Vlan20	EDITORIAL	172.30.84.33	255.255.255.224
Vlan30	SALES	172.30.84.65	255.255.255.224
Vlan40	DESIGN	172.30.84.97	255.255.255.224
Vlan50	BOARDROOM	172.30.84.129	255.255.255.240
Vlan60	SERVERS	172.30.84.145	255.255.255.248
Vlan70	VOICE reserve	172.30.84.161	255.255.255.224
Vlan99	MANAGEMENT	172.30.84.193	255.255.255.240

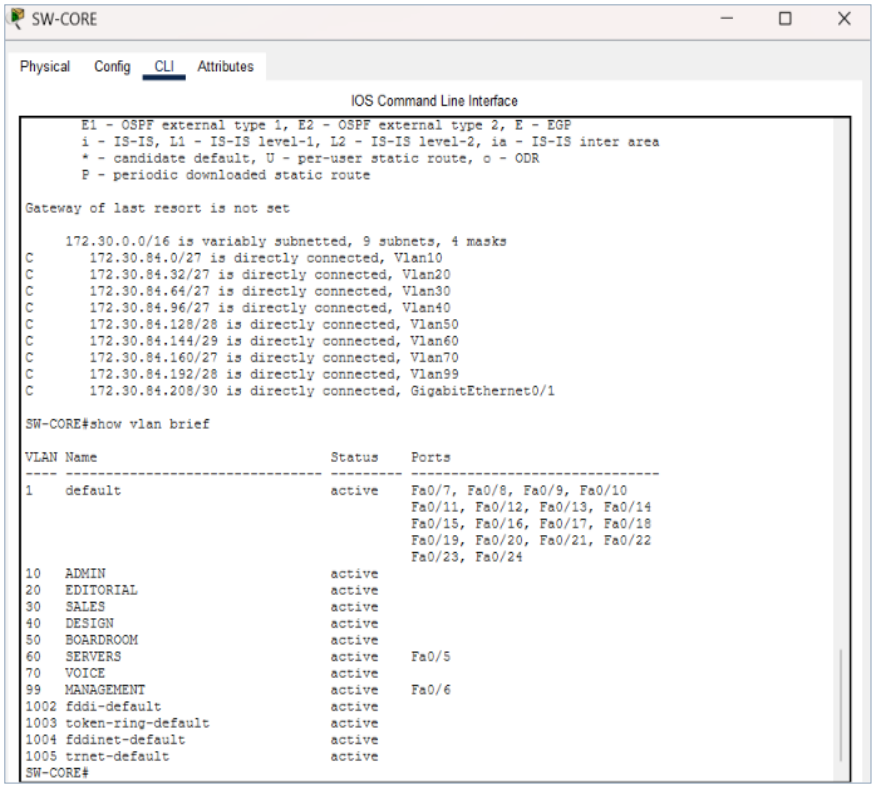


Figure 2 — VLAN list on SW-CORE (`show vlan brief`)

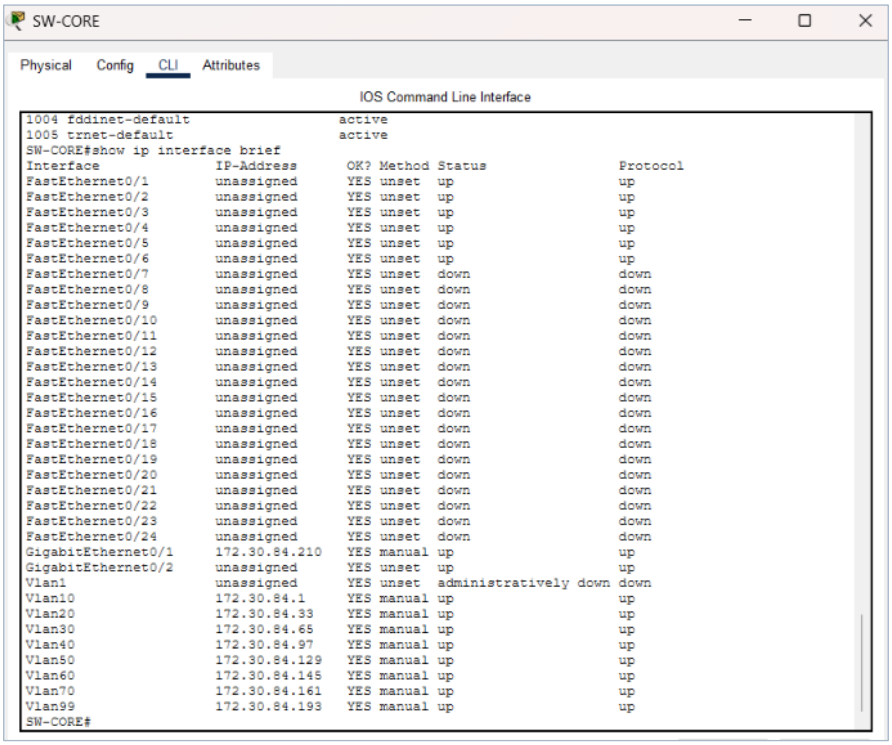


Figure 3 — SVI status on SW-CORE (`show ip interface brief`)

3.6 Management Plane Addressing

VLAN 99 uses 172.30.84.192/28. PC-MANAGEMENT is 172.30.84.194, SW-CORE provides gateway 172.30.84.193, and the access switches have unique addresses. R1 is reached using 172.30.84.209 on its internal /30 connection rather than through a VLAN 99 SVI.

Table 5 — Management plane IP addresses

Device	Management IP
SW-CORE (VLAN 99)	172.30.84.193
PC-MANAGEMENT	172.30.84.194
SW-ADMIN	172.30.84.195
SW-EDIT	172.30.84.196
SW-SALES	172.30.84.197
SW-DESIGN	172.30.84.198
SW-BOARD	172.30.84.199
R1 (internal boundary link)	172.30.84.209

3.7 Router Configuration (R1)

R1 is connected to SW-CORE on GigabitEthernet0/0 over 172.30.84.208/30. It uses static routes to reach the VLAN 99 management network through SW-CORE. GigabitEthernet0/1 and GigabitEthernet0/2 are shut down, and the project does not implement an external WAN or Internet connection.

Table 6 — R1 configuration summary

Item	Configuration
Gig0/0	172.30.84.209/30 toward SW-CORE 172.30.84.210
Management subnet route	ip route 172.30.84.192 255.255.255.240 172.30.84.210
Default route in saved implementation	ip route 0.0.0.0 0.0.0.0 172.30.84.210
Gig0/1 and Gig0/2	Administratively down; no ISP uplinks

4. Assigned Networking Feature: SSH

4.1 Design Approach

The assigned challenge was to implement secure remote device management using Secure Shell (SSH). SSH version 2 was configured so that administration takes place through authenticated, encrypted sessions instead of Telnet. The management workstation resides in VLAN 99 and connects to SW-CORE and the access switches using their management addresses. R1 is accessed via its internal point-to-point interface.

4.2 Configuration Applied

The implementation uses a local administrator account, domain name, RSA keys, SSH version 2 and local VTY authentication. During the final corrections, the five access switches were configured so that their VTY lines 0–15 use `login local` and `transport input ssh`. The router and core switch use SSH-only settings on their implemented VTY lines 0–4. The sample below uses a placeholder instead of disclosing an administrative secret.

```
ip domain-name molemane.local
username admin privilege 15 secret [REDACTED – NOT FOR PUBLICATION]
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
! On access switches:
line vty 0 15
 login local
 transport input ssh
! R1 and SW-CORE: SSH-only on their configured line vty 0 4
```

4.3 Key Decisions

- SSH version 2: used for secure remote management in place of Telnet.
- Local login: the administrator account enables device-level authentication without a separate AAA server.
- RSA key generation: 2048-bit keys were configured as part of SSH setup.
- VLAN 99: separates management addressing from the departmental user VLANs and is the configured native VLAN on trunks.
- VTY coverage: the access-switch lines 0–15 was corrected in the final working file to prevent unprotected extra VTY lines.
- Internal-only design: SW-CORE handles inter-VLAN routing while R1 has no external ISP connection.

4.4 Verification

The `show ip ssh` command on SW-CORE confirmed SSH version 2.0. From PC-MANAGEMENT, remote SSH logins were successfully established to R1 and all six switches in the final checks. After login, `show privilege` confirmed privilege level 15. This demonstrates access to the management CLI, not a separate audit of every possible privileged command.

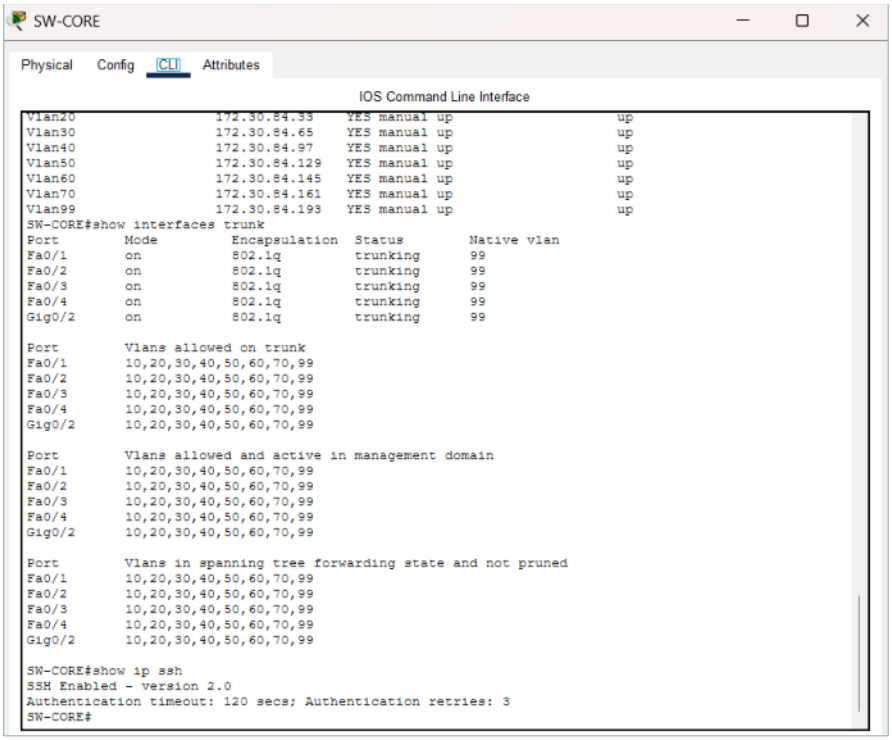


Figure 4 — SSH version 2 enabled on SW-CORE (`show ip ssh`)

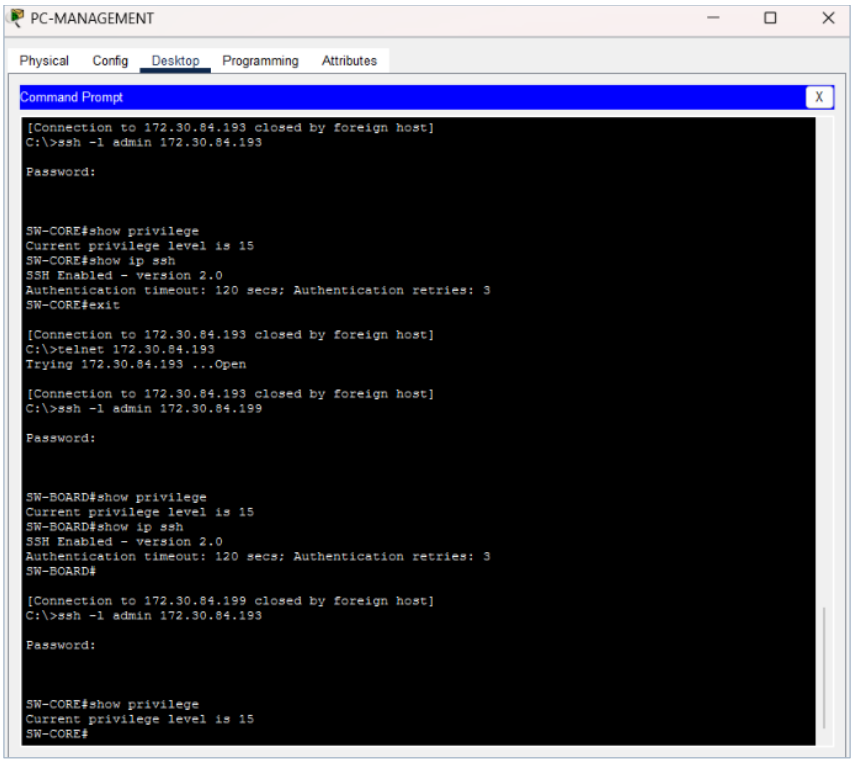


Figure 5 — Successful SSH login from PC-MANAGEMENT to SW-CORE

4.5 Telnet Rejection

Telnet was tested from PC-MANAGEMENT to 172.30.84.193 and to the management addresses of other devices. The connection was closed by the remote host without a usable login session. A brief “Open” message before disconnection is not evidence of a successful Telnet login. These observations support the SSH-only VTY configuration.



Figure 6 — Telnet rejection from PC-MANAGEMENT

5. Testing and Verification

5.1 Inter-VLAN Connectivity Tests

The original Milestone 2 ping tests generally returned three replies out of four on the first attempt. During final testing, most destinations responded to all four packets on a repeated test. PC-DESIGN remained reachable, but one packet timed out in repeated four-packet tests. The results below record the verified final-stage outputs; they should not be represented as application-level service tests.

Table 7 — Final IP connectivity checks

Source	Destination (IP)	VLAN	Packets received	Result
PC-ADMIN	PC-EDIT (.34)	20	4 / 4 on retest	Reachable; 0% loss
PC-ADMIN	PC-SALES (.66)	30	4 / 4 on retest	Reachable; 0% loss
PC-ADMIN	PC-DESIGN (.98)	40	3 / 4 on repeated test	Reachable; 25% loss
PC-ADMIN	Wired boardroom (.130)	50	4 / 4 on retest	Reachable; 0% loss
PC-ADMIN	Server (.146)	60	4 / 4 on retest	Reachable; 0% loss
Wireless laptop (.131)	VLAN 50 gateway (.129)	50	4 / 4	Reachable; 0% loss
Wireless laptop (.131)	Server (.146)	60	4 / 4 after WPA2	Reachable; 0% loss

Wireless verification was performed on BOARDROOM-WIRELESS-LAPTOP after installing the WPC300N module and associating BOARDROOM-WIRELESS-AP. The laptop was statically configured as 172.30.84.131/28 with default gateway 172.30.84.129. It received four replies when pinging both the gateway and the server. After the access point was secured using WPA2-PSK and the laptop reconnected, another server ping received four out of four replies.

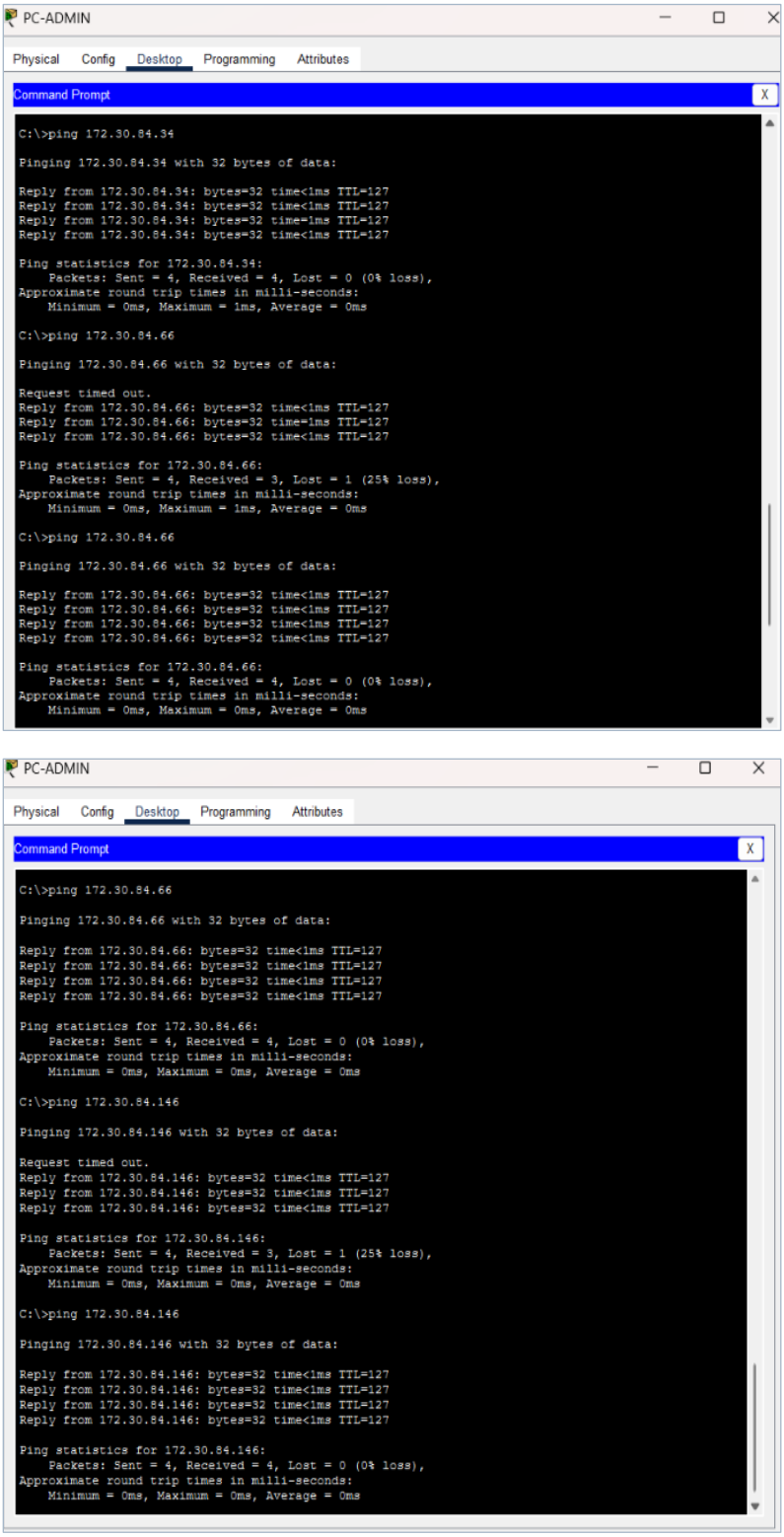


Figure 7 — Inter-VLAN ping tests from PC-ADMIN (parts 1 and 2)

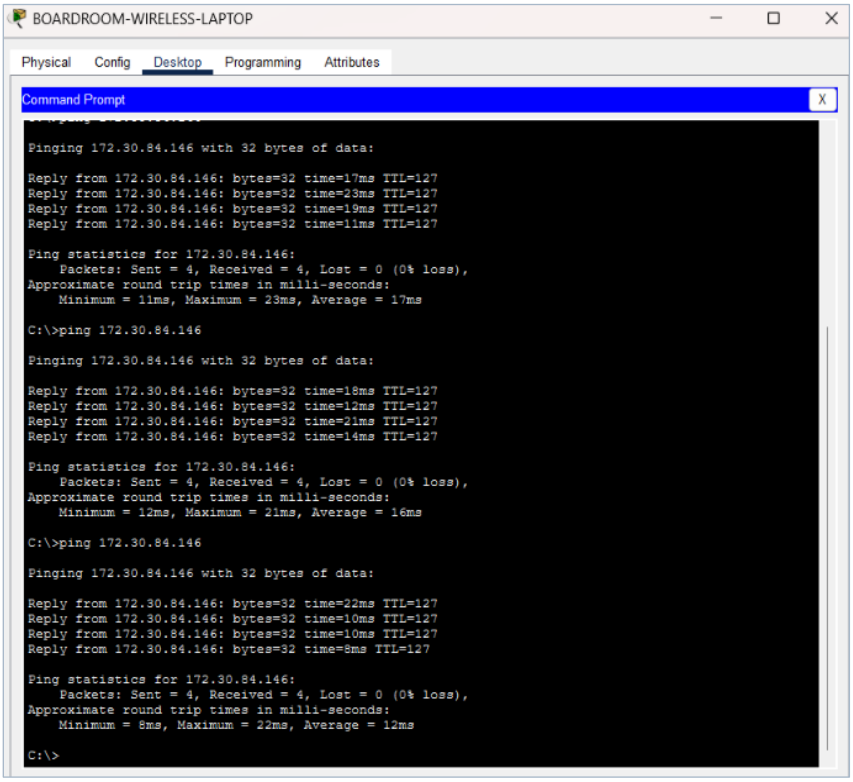


Figure 8 — Boardroom wireless laptop pinging the server (172.30.84.146)

5.2 SSH Tests

All seven network devices were tested from PC-MANAGEMENT during final verification. After successful SSH login, `show privilege` showed privilege level 15. Telnet rejection was also observed. The results below distinguish the current final checks from the earlier Milestone 2 screenshots.

Table 8 — Final SSH verification

Device	SSH destination	SSH login	Privilege
R1	172.30.84.209	Successful	15
SW-CORE	172.30.84.193	Successful	15
SW-ADMIN	172.30.84.195	Successful	15
SW-EDIT	172.30.84.196	Successful	15
SW-SALES	172.30.84.197	Successful	15
SW-DESIGN	172.30.84.198	Successful	15
SW-BOARD	172.30.84.199	Successful	15

SW-CORE additionally reported “SSH Enabled - version 2.0” using `show ip ssh`. Telnet sessions were closed without granting administrator access. Existing Milestone 2 evidence is retained in the portfolio, with final-stage results documented separately.

5.3 Trunk Tests

In Milestone 2, the `show interfaces trunk` command on SW-CORE verified trunk ports, native VLAN 99, the 802.1Q encapsulation and allowed VLANs. The completed network retained this trunk design. Figure 9 shows the trunk status captured from the final working Packet Tracer file, with all five trunk interfaces operating in 802.1Q mode, native VLAN 99 and VLANs 10, 20, 30, 40, 50, 60, 70 and 99 allowed.

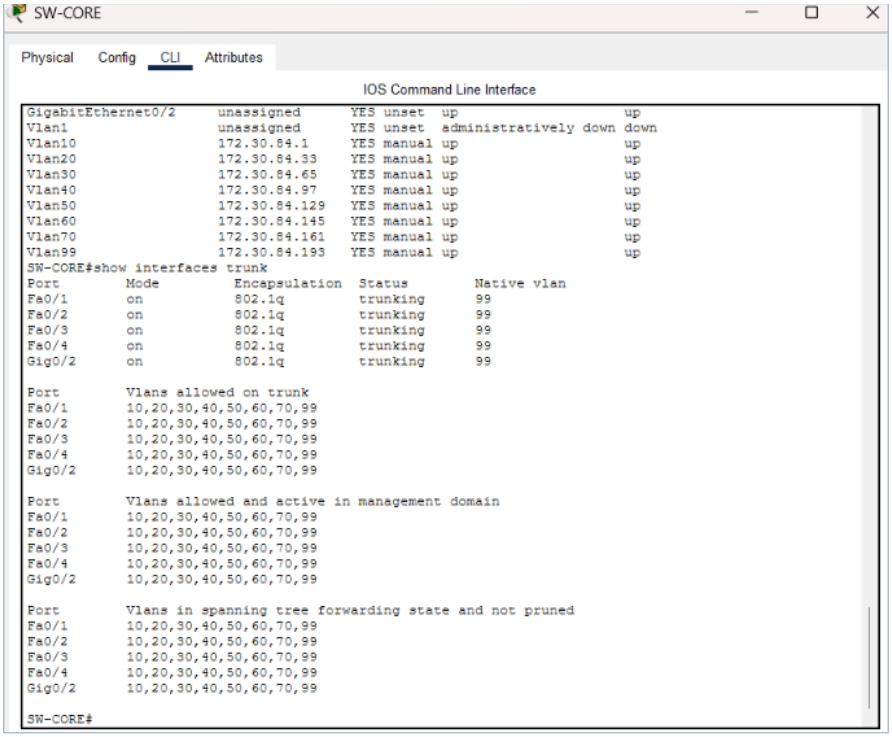


Figure 9 — Trunk status on SW-CORE (`show interfaces trunk`)

6. Interpretation of Results

The completed network demonstrates IP reachability between VLAN 10 and the other tested departmental, boardroom and server networks. Most destinations returned four of four replies on a repeated ping, while PC-DESIGN returned three of four in repeated tests. The latter still demonstrates reachability but should not be described as 0% loss. Initial Packet Tracer address resolution is one possible cause of first-packet loss, although it was not independently isolated as the cause of every timeout.

The secured wireless laptop reached the VLAN 50 gateway and the server in VLAN 60, indicating that the access point, boardroom VLAN connection, SVI gateway and routed path to the server were functioning in those tests. WPA2-PSK was enabled and the laptop reconnected successfully. These are connectivity and association results; they do not demonstrate that the internal file, print or DNS applications themselves were exercised.

The SSH tests confirm that administrative connections were established from PC-MANAGEMENT and that privilege level 15 was reported on all seven devices. Telnet login was rejected. The final VTY corrections are reflected in the updated working Packet Tracer file, while historical Milestone 2 configuration exports remain in the GitHub portfolio for comparison.

The reserved VLAN 70 has an assigned SVI on SW-CORE, but no VoIP handsets are included. Its configuration is therefore a readiness provision for CR11, not evidence of implemented voice calls. The network does not include an external ISP or Internet service.

7. Troubleshooting Log

The first nine issues below were documented during Milestone 2. The final two entries record subsequent management-security and boardroom-wireless improvements. These entries explain the change from the earlier configuration snapshots to the final working implementation.

Table 9 — Issues identified and resolved or improved

#	Issue	Cause / observation	Correction or action
1	Native VLAN mismatch	Trunk endpoints had different native VLANs.	Set VLAN 99 as native on both ends.
2	Missing SVIs on SW-CORE	Initial SVI configuration stopped at VLAN 40.	Added SVIs for VLANs 50, 60, 70 and 99.
3	Server ping initially failed	VLAN 60 SVI was missing.	Configured Vlan60, gateway 172.30.84.145/29.
4	R1 SSH timeout	R1 lacked a route to VLAN 99.	Added static route via 172.30.84.210.
5–7	Duplicate gateway IPs	Access switches had data-VLAN SVIs using core gateway addresses.	Removed / shut down the extra data-VLAN SVIs.
8	SW-BOARD port in wrong VLAN	Port not reassigned after cabling adjustment.	Assigned relevant access port to VLAN 50.
9	Management IP conflict	SW-CORE and PC-MANAGEMENT initially used .194.	SW-CORE .193; PC-MANAGEMENT .194.
10	VTY coverage incomplete	Older access-switch exports secured only VTY 0–4; VTY 5–15 showed login.	Corrected VTY 0–15 to local login and SSH-only; retested remote SSH and Telnet.
11	Boardroom Wi-Fi validation / security	Existing AP lacked a separate tested wireless client; security was initially disabled.	Added wireless laptop, configured VLAN 50 IP, enabled WPA2-PSK and retested server ping.

8. Conclusion

The completed Packet Tracer implementation follows the main Milestone 1 network design and improves upon the Milestone 2 submission. VLANs 10–40 separate the newspaper departments, VLAN 50 supports the boardroom, VLAN 60 contains the internal server, VLAN 99 supports device management and VLAN 70 remains reserved for future CR11 voice devices. SW-CORE provides the inter-VLAN gateways, and R1 remains an internal router without an ISP connection.

Final network tests confirmed communication across the tested VLANs, successful connection of a WPA2-PSK-secured wireless laptop, and IP reachability from that laptop to the server. The assigned SSH challenge was retested across R1 and all six switches, with administrator privilege level 15 and rejection of Telnet logins. The final corrections addressed VTY coverage and provided stronger evidence of secure boardroom wireless access.

The work highlighted the importance of consistent trunk settings, avoiding duplicate SVI gateway addresses, planning management reachability, and verifying results with actual ping and IOS command output. The corrected Packet Tracer file was uploaded separately to the GitHub portfolio while original milestone materials were preserved. The final submission comprises the Packet Tracer file, the GitHub portfolio, this technical report and the demonstration video.

9. References

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